WASP-43b

R-0450

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_240305 · created 2026-07-19T16:42:25+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460374.7347 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.015
Transit depth (Rp/Rs)^2	3.84 +/- 0.58 [%]
Orbital Inclination (inc)	80.31 +/- 1.21
Airmass coefficient 1 (a1)	0.992 +/- 0.011
Airmass coefficient 2 (a2)	0.004 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#2 - [400, 53]
Optimal Aperture	3.78
Optimal Annulus	5.22
Transit Duration (day)	0.0467 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.015 vs 0.1594 (deviation 2.44σ)
Duration fit vs published	0.0467 d vs 0.0512 d (deviation 0.86σ)
Mid-time O–C	1.4 min
Depth SNR	13.1
Residual scatter	1.12 %

Light curve

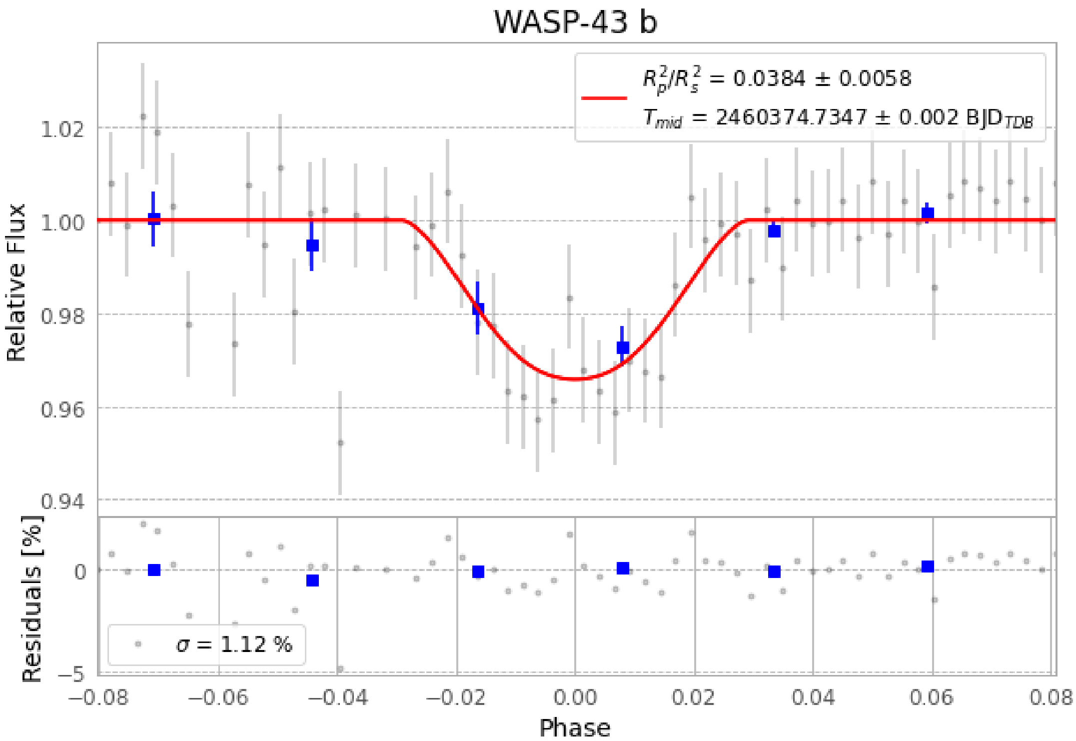

Figure 1 — FinalLightCurve_WASP-43 b_2024-03-04.png (SHA-256 2b4ca6f6e3d447a6…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [487, 223], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 808273543ac6b617…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

64 FITS frames (42 MB), `WASP-43240305035415.FITS` → `WASP-43240305070315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240305.log (a22856b748a0…), FinalLightCurve_WASP-43 b_2024-03-04.png (2b4ca6f6e3d4…), FinalLightCurve_WASP-43 b_2024-03-04.csv (ee3e37a98754…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)